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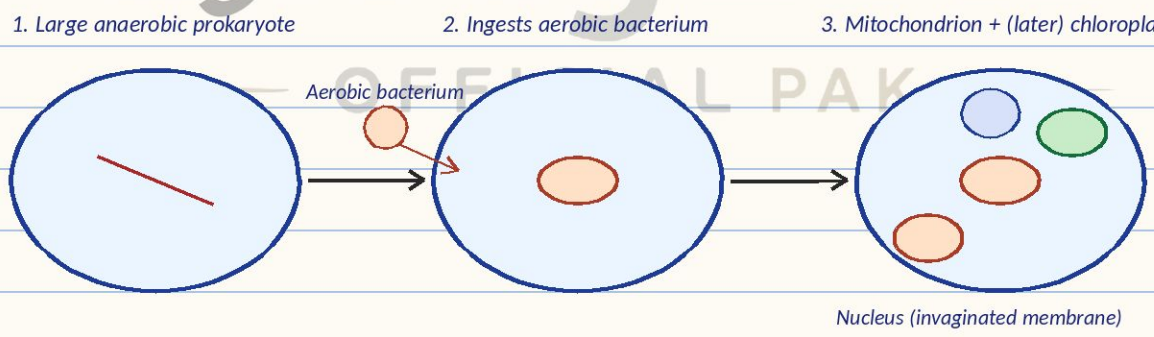
Chapter 14: Evolution

Evolution: Change over a period of time; the gradual change in living organisms through generations, giving diversity of life.

- Earth is about **4.5 billion years** old; the oldest known fossils are **3.7 billion years** old. The first organisms were **prokaryotic heterotrophs** feeding on organic molecules of the ocean "soup". As the soup ran short, some became **autotrophs** able to use solar energy.
- The earliest photosynthetic autotrophs used **hydrogen sulphide** as hydrogen source (no oxygen). Later cyanobacteria-like organisms used **water** and released oxygen; rising oxygen formed the **ozone layer**, which cut UV radiation and made life on land possible.

14.1 Evolution from prokaryotes to eukaryotes (SLO 5)

- Prokaryotes ruled the world for over a billion years; eukaryotes appeared **1.9 to 2.1 billion years ago** (fossils). Most striking theory: **endosymbiosis** proposed by **Lynn Margulis.**



Evidence: own circular DNA, 70S ribosomes, double membrane, binary fission, size of bacteria
Spirochete ingestion may have given flagella / cilia; photosynthetic bacterium gave chloroplasts

Fig: Endosymbiotic origin of eukaryotes (Lynn Margulis)

- A large **anaerobic amoeboid prokaryote** ingested a small **aerobic bacterium**

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*and kept it alive as an **endosymbiont**; the bacterium became the **mitochondrion** (aerobic respiration advantage). Flagella may have come from ingested **spirochete-like** prokaryotes. **Chloroplasts** came from photosynthetic bacteria (cyanobacteria) taken in by a second endosymbiotic event.*

- ***Evidence:** mitochondria and chloroplasts are similar in size to bacteria, have their **own DNA** (circular), **bacteria-like ribosomes**, make some of their own proteins and divide by fission; they have two membranes. Over time endosymbionts transferred many genes to the host nucleus and lost independence.*
- *Eukaryotes brought a dramatic increase in complexity and diversity: independent single cells, later **multicellular** organisms with specialized cells.*

14.2 Lamarckism (SLO 6, 7)

- ***Jean-Baptiste de Lamarck** was the first to put forward a theory of evolution (1809, **Philosophie Zoologique**). Two ideas: **use and disuse of organs** and **inheritance of acquired characters**.*
- ***Use and disuse:** frequently used organs develop and become strong (blacksmith's biceps; webbed feet of aquatic birds); unused organs degenerate and finally disappear (snakes lost legs through creeping; human ear muscles lost contractility).*
- ***Inheritance of acquired characters:** characters acquired during an individual's lifetime are passed to offspring, accumulate through generations and fit the group better to its environment.*
- ***Giraffe neck:** short-necked ancestors fed on leaves at progressively higher levels; each stretched its neck to reach the top leaves, each offspring was born with a slightly longer neck, and so on until the present long-necked giraffe.*

Drawbacks of Lamarckism

1. *No experimental proof of Lamarckism.*
2. *Use and disuse of organs cannot alter the genotype of the organism.*
3. *Somatic (body) changes are not heritable; acquired traits are non-genetic.*

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4. *Organs are not modified by the wish or requirement of the organism.*

14.3 Darwinism (SLO 1)

- ***Charles Darwin*** *(born in England into a wealthy family; studied theology at Cambridge but attended biology and geology lectures) sailed at 22 in 1831 as naturalist on* ***HMS Beagle*** *around the world. At the* ***Galapagos Islands*** *(965 km west of Ecuador) he made his most important observations.*
- ***South American mainland:*** *flora and fauna have a distinct South American stamp unlike Europe; South American fossils, though different from modern species, resemble the living species of the continent.*
- ***Galapagos:*** *species resemble those of the mainland more than those of other islands, yet differ; closely related* ***finches*** *(14 types) had beaks of different sizes and shapes adapted to different foods. Some were unique to an island, some found on neighbouring islands.*

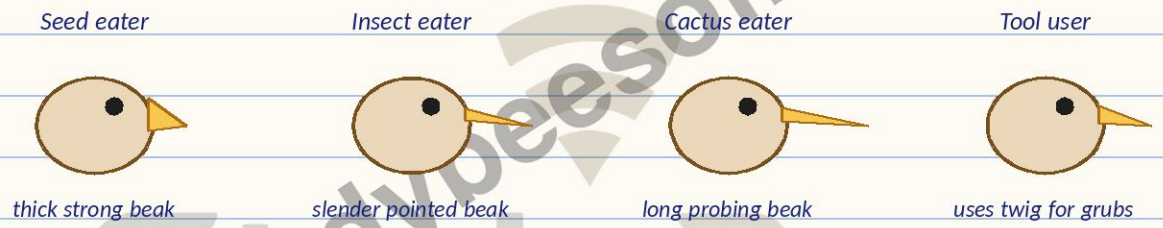


Fig: Darwin's finches (adaptive radiation)

- *Darwin saw origin of new species and adaptation as related: a new species arises from an ancestral form by gradual accumulation of adaptations to different environments separated by geographical barriers.*
- ***Development of the theory:*** *began late 1830s; wrote a 35-page sketch (1842) and a 230-page essay (1844); gathered facts for 15 more years. Influences:*

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Scientist	Contribution
Charles Lyell and James Hutton	Lyell's **Principles of Geology** supported Hutton's **uniformitarianism**: slow geological forces still operating shaped earth, so earth is very old.
Thomas Malthus	**Essay on the Principle of Population**: human population increases exponentially (1, 2, 4, 8, 16) but food supply arithmetically (1, 2, 3, 4): struggle for existence.
Alfred Russel Wallace	In 1858 sent Darwin an essay with the same theory; papers presented jointly to the Linnaean Society on 1 July 1858.

- **The Origin of Species by Means of Natural Selection** appeared in November 1859; **Descent of Man** 12 years later. The theory is attributed to Darwin because he explored its implications (especially for humans), while Wallace embraced radical social causes and spiritualism.

Darwin's theory of natural selection (SLO 1)

- Two main points: (i) **descent with modification** (all organisms are related by descent from common ancestors: history of life is a branching tree); (ii) **natural selection and adaptation**.

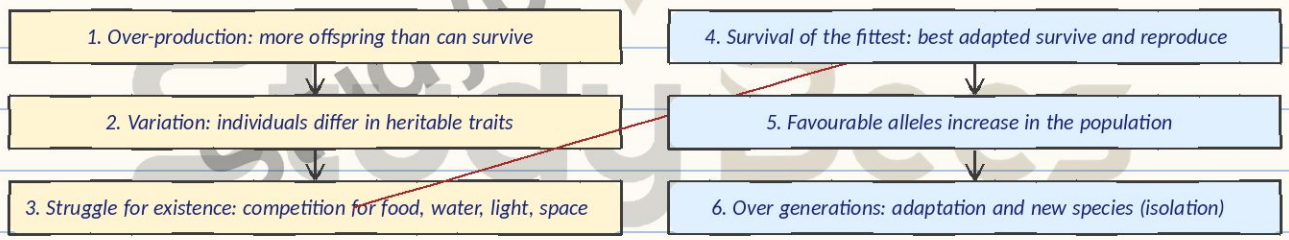


Fig: Darwin's natural selection (steps)

- **Natural selection:** differential reproductive capacity among the individuals of a population. Four observations: **over-production**, **variations** (some improve survival and reproduction), **struggle for existence** (interspecific, intraspecific and environmental; also predators and disease) and **survival of the fittest**

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(fittest = those that reproduce most successfully; e.g. in a flood swimmers survive, in an earthquake flyers).

- *It increases favourable alleles and decreases unfavourable ones; members become better adapted to local conditions, leading to new species.* ***Selection pressure*** *= environmental factors (climate, predators, food shortage, disease) that favour some variants.*

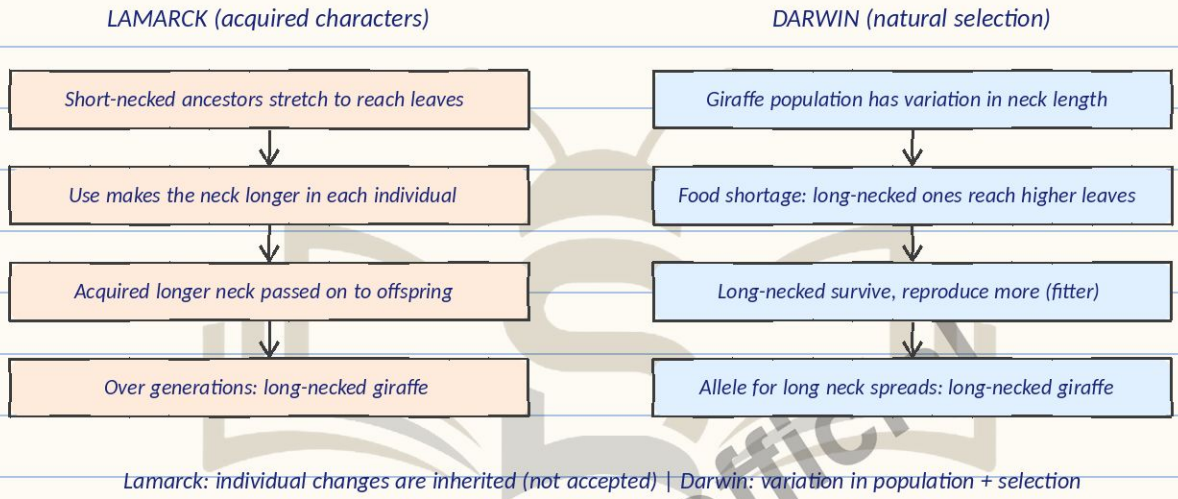


Fig: Evolution of giraffe neck - Lamarck and Darwin

Lamarckism	***Darwinism***
Inheritance of acquired characters	*Natural selection of heritable variations*
Changes arise from need and use of organs	*Variations arise randomly; environment selects*
Individual adapts and passes on change	*Population changes as favourable alleles increase*
Not supported by genetics	*Supported by modern genetics (neo-Darwinism)*

14.4 Neo-Darwinism

- *Lamarck and Darwin knew nothing of heredity.* ***Population genetics*** *now explains inheritance and variation and strongly supports Darwin; this reappraisal of natural selection in terms of modern population genetics =* ***Neo-Darwinism*** *(variation from mutation, crossing over, random assortment; selection acts on the phenotypes and changes allele frequencies; differential*

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reproduction).

14.5 Evidences in favour of evolution

- *Evidence comes from biogeography, molecular biology, comparative anatomy, comparative embryology, palaeontology (fossils), animal and plant breeding and taxonomy.*

Biogeography (SLO 2)

***Biogeography:** The study of geographical distribution of plants and animals on Earth. It gives evidence of prehistoric climates, habitats and animal distribution patterns.*

- *Life-forms in different regions have distinct evolutionary histories. Species **originate in a particular area then disperse**; absence of a species from an area indicates earlier separation of that area from the region of origin. **Continental drift** (continents moving over 225 million years) separated land masses. Geographical distribution is what first suggested evolution to Darwin.*
- ***Specific pattern of distribution:** South America lacks rabbits though suitable, because rabbits originated elsewhere and could not reach it; rabbits are not native to Australia, but when introduced increased rapidly (suitable environment).*
- ***Factors inhibiting distribution (barriers): physical** (ocean, desert, mountain), **environmental** (unfavourable climate) and **ecological** (competitors for food and shelter).*

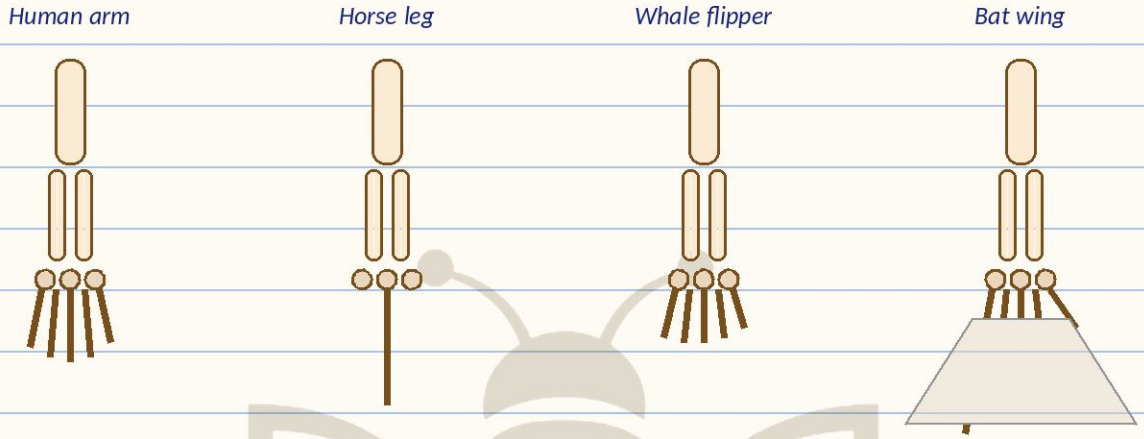
Molecular biology (SLO 3)

- *Similar molecules in all organisms show **biochemical homology** like anatomical homology. **DNA** (same nucleotides) is the hereditary material of all organisms with a **universal genetic code**; **ATP** is the energy currency from prokaryotes to advanced plants and animals; proteins from the same **20 amino acids**; **cytochrome c** is in the respiratory chain of all aerobic organisms (differences in its sequence reflect relationship). These show origin from a common ancestor.*

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Comparative anatomy (SLO 4)

- *Study of similarities and differences in structures of different species. Similar body parts may be **homologies** or **analogies**; both are evidence.*



Humerus - radius + ulna - carpals - metacarpals - phalanges (same bone pattern, different functions)

Homologous structures = divergent evolution from a common ancestor

Fig: Homologous forelimbs of mammals (common bone plan)

- ***Homologous structures:** look different (different functions) but are **internally alike** due to **common origin**; e.g. arm of man, forelimb of horse, flipper of whale, wing of bat (same bones); sepals, petals, stamens and carpels (modified leaves); **vestigial organs** (appendix, tail vertebrae, ear muscles in humans; leg bones in some snakes). Pattern = **divergent evolution**: different species from a common ancestor in different habitats.*



Same function (flight), different structure and origin = analogous structures

Convergent evolution: unrelated species in a similar habitat evolve similar structures

Fig: Analogous wings of bird, bat and insect

- ***Analogous structures:** look alike because they do the same job but have*

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different internal organization and origin; *e.g. wings of bats, birds and insects; leaves of tracheophytes, mosses and algal blades. Pattern =* ***convergent evolution***: *species from different ancestors in a common habitat. They show that populations with separate ancestries adapt in similar ways to similar demands.*

Homologous (divergent evolution)	***Analogous (convergent evolution)***
Same origin, similar basic structure	*Different origin, different structure*
Different functions	*Same function*
Related species, different habitats	*Unrelated species, same habitat*
Arm of man, bat wing, whale flipper	*Wings of bird and insect*

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Chapter 14: Short Questions with Answers (SLO based)

Q1. [SLO 1] Define evolution.

Ans: Change over a period of time; gradual change in living organisms through generations.

Q2. [SLO 1] What was the first life form on earth?

Ans: Prokaryotic heterotrophs feeding on the organic molecules of the ocean "soup".

Q3. [SLO 1] How did photosynthesis change life on earth?

Ans: Oxygen-evolving photosynthesis raised oxygen and formed an ozone layer that reduced UV, allowing life on land.

Q4. [SLO 1] What is variation? How does it lead to evolution?

Ans: Differences between individuals of a population; if favourable to the environment they raise survival and reproduction and spread.

Q5. [SLO 1] What is selection pressure?

Ans: Environmental factors such as climate, predators, disease and food shortage that favour some variants over others.

Q6. [SLO 1] List the four observations of Darwin's natural selection.

Ans: Over-production, variations, struggle for existence and survival of the fittest.

Q7. [SLO 1] Define natural selection.

Ans: Differential reproductive capacity among individuals of a population: those with favourable traits survive and reproduce more.

Q8. [SLO 1] Natural selection is random or systematic? Explain.

Ans: Variations arise randomly, but selection is systematic: the environment consistently favours the best adapted, increasing favourable alleles.

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Q9. [SLO 1] What is meant by survival of the fittest?

Ans: The fittest are the individuals that reproduce most successfully in the environment, not simply the strongest, cleverest or most intelligent.

Q10. [SLO 1] What is descent with modification?

Ans: All organisms are related through descent from common ancestors, and adaptation to environments gives diversity (tree of life).

Q11. [SLO 1] Name the books of Darwin.

Ans: The Origin of Species by Means of Natural Selection (1859) and The Descent of Man.

Q12. [SLO 1] What did Darwin observe about finches in the Galapagos?

Ans: 14 similar species with beaks of different sizes and shapes adapted to different foods, from one mainland ancestor.

Q13. [SLO 1] What was the contribution of Lyell, Malthus and Wallace to Darwinism?

Ans: Lyell: very old earth by slow geological forces; Malthus: population outgrows food supply so struggle; Wallace: independently proposed natural selection (1858).

Q14. [SLO 1] Why is the theory attributed to Darwin and not Wallace?

Ans: Darwin explored its implications, especially for humans, and published a full book; Wallace embraced radical social causes and spiritualism.

Q15. [SLO 1] What is neo-Darwinism?

Ans: Reappraisal of natural selection in the light of modern population genetics.

Q16. [SLO 1] What is uniformitarianism?

Ans: Hutton's theory (supported by Lyell) that the same slow geological processes occurring today shaped the earth over a very long time.

Q17. [SLO 2] Define biogeography.

Ans: Study of the geographical distribution of plants and animals on Earth.

Q18. [SLO 2] How does biogeography support evolution?

Ans: Species originate in an area and disperse; absence of species shows earlier separation; similar species on mainland and islands show common ancestry.

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Q19. [SLO 2] Why are there no rabbits in South America?

Ans: Rabbits originated elsewhere and could not reach the continent, though the environment suits them.

Q20. [SLO 2] Name barriers that restrict dispersal of organisms.

Ans: Physical (ocean, desert, mountain), environmental (unfavourable climate) and ecological (competitors).

Q21. [SLO 2] What is continental drift?

Ans: Slow movement of the continents; over the last 225 million years it separated land masses and isolated species.

Q22. [SLO 3] What is biochemical homology?

Ans: Similarity of molecules (DNA, ATP, cytochrome c, amino acids) among organisms showing common ancestry.

Q23. [SLO 3] How does cytochrome c support evolution?

Ans: It is present in the respiratory chain of all aerobic organisms; its similar sequence reflects common ancestry.

Q24. [SLO 3] Name common biomolecules in all living organisms.

Ans: DNA with universal genetic code, ATP, 20 amino acids and cytochrome c.

Q25. [SLO 4] Define homologous and analogous structures.

Ans: Homologous: same origin and basic structure, different function. Analogous: same function, different origin and structure.

Q26. [SLO 4] What are vestigial organs? Give human examples.

Ans: Reduced, non-functional organs homologous to useful organs of ancestors: appendix, tail vertebrae, ear muscles, wisdom teeth.

Q27. [SLO 4] Differentiate convergent and divergent evolution.

Ans: Divergent: different species from a common ancestor in different habitats (homologous structures). Convergent: different ancestors in a common habitat evolve similar structures (analogous).

Q28. [SLO 4] Give two examples of homologous and analogous structures.

Ans: Homologous: human arm and bat wing; sepals and petals. Analogous: wings of bird and insect; leaves of tracheophytes and moss.

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Q29. [SLO 4] Are the wings of bird and bat homologous or analogous?

Ans: The wing as a flying structure is analogous (different origin); the forelimb bones of bat and bird are homologous.

Q30. [SLO 4] List few characteristics found only in mammals.

Ans: Hair, mammary glands, diaphragm, three ear bones, pinna, non-nucleated RBCs.

Q31. [SLO 5] What is the endosymbiotic theory? Who proposed it?

Ans: Proposed by Lynn Margulis: mitochondria and chloroplasts originated from aerobic and photosynthetic bacteria ingested by a prokaryote that kept them as endosymbionts.

Q32. [SLO 5] Give evidence for endosymbiosis.

Ans: Mitochondria and chloroplasts have own circular DNA, 70S ribosomes, bacterial size, double membranes and divide by fission.

Q33. [SLO 5] How did mitochondria and flagella arise according to this theory?

Ans: Mitochondria: from aerobic bacteria ingested by an anaerobic cell. Flagella: from ingested spirochete-like prokaryotes.

Q34. [SLO 5] When did eukaryotes appear?

Ans: About 1.9 to 2.1 billion years ago.

Q35. [SLO 6] State Lamarck's theory.

Ans: Use and disuse of organs and inheritance of acquired characters (Philosophie Zoologique, 1809).

Q36. [SLO 6] What are acquired characters? Give examples.

Ans: Characters gained during lifetime by use and disuse of organs, e.g. blacksmith's biceps, webbed feet of aquatic birds.

Q37. [SLO 6] Explain the evolution of giraffe neck according to Lamarck.

Ans: Short-necked ancestors stretched to reach high leaves; the longer neck acquired was inherited, continuing until the present neck length.

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Q38. [SLO 7] State drawbacks of Lamarckism.

Ans: No experimental proof; use and disuse cannot change genotype; somatic changes are not heritable; organs do not change by wish.

Q39. [SLO 7] Why are acquired characters not inherited?

Ans: They are changes in body (somatic) cells that do not alter the genes in gametes.

Q40. [SLO 7] Differentiate between Lamarckism and Darwinism.

Ans: Lamarck: acquired characters inherited. Darwin: heritable variations selected by environment.

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Chapter 14: Long Questions with Answers (SLO based)

Q1. [SLO 1] How does natural selection operate to create new species? Discuss in detail.

Ans:

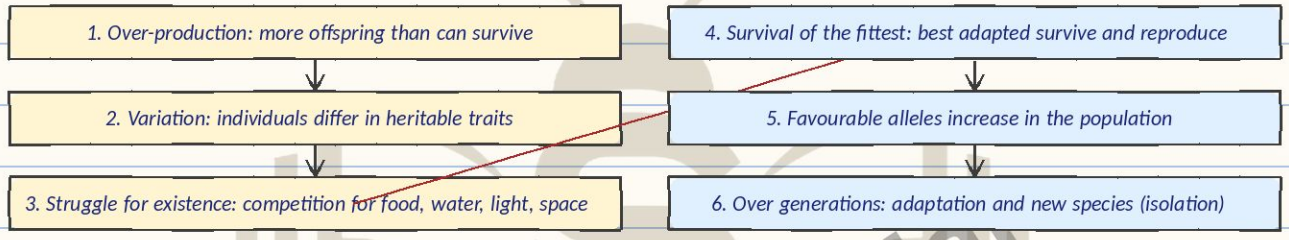


Fig: Darwin's natural selection (steps)

- *Over-production, variations, struggle for existence (interspecific, intraspecific, environmental), survival of the fittest; favourable alleles increase; with geographical barriers populations adapt to different environments (Galapagos finches) and over many generations become separate species.*
- *Neo-Darwinism: variation from mutation and recombination; selection changes allele frequencies; differential reproduction.*

Q2. [SLO 3] Explain evidences from molecular biology in support of evolution.

Ans:

- *Biochemical homology: DNA is the hereditary material of all organisms with a universal genetic code; ATP is the common energy currency; proteins are made of the same 20 amino acids; cytochrome c is found in the respiratory chain of*

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all aerobic organisms.

- *Such common molecules and roles reflect evolutionary relationship and origin from a common ancestor.*

Q3. [SLO 2] Suppose a new mammalian species originated in the plains of the sub-continent. What kind of barriers can restrict its dispersal in other regions of the world?

Ans:

- *Physical barriers: oceans (Arabian Sea, Indian Ocean), deserts, and high mountains (Himalaya, Hindu Kush).*
- *Environmental barriers: unfavourable climate (extreme cold or heat, rainfall).*
- *Ecological barriers: competitors for food and shelter, predators and disease organisms in new regions.*

Q4. [SLO 2] Discuss the evidence provided by biogeography.

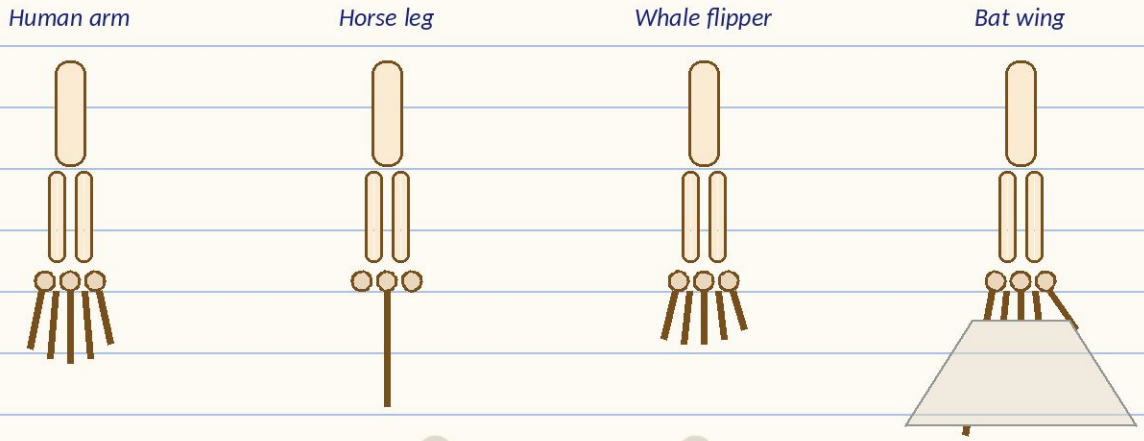
Ans:

- *Distribution of species and fossils inspired Darwin: South American flora and fauna; Galapagos species resemble mainland species; rabbits absent from South America and Australia until introduced.*
- *Species originate and disperse; barriers (physical, environmental, ecological) limit distribution; continental drift separated land masses over 225 million years.*

Q5. [SLO 4] Differentiate convergent and divergent evolution on the basis of homologous and analogous structures.

Ans:

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Humerus - radius + ulna - carpals - metacarpals - phalanges (same bone pattern, different functions)

Homologous structures = divergent evolution from a common ancestor

Fig: Homologous forelimbs of mammals (common bone plan)



Same function (flight), different structure and origin = analogous structures

Convergent evolution: unrelated species in a similar habitat evolve similar structures

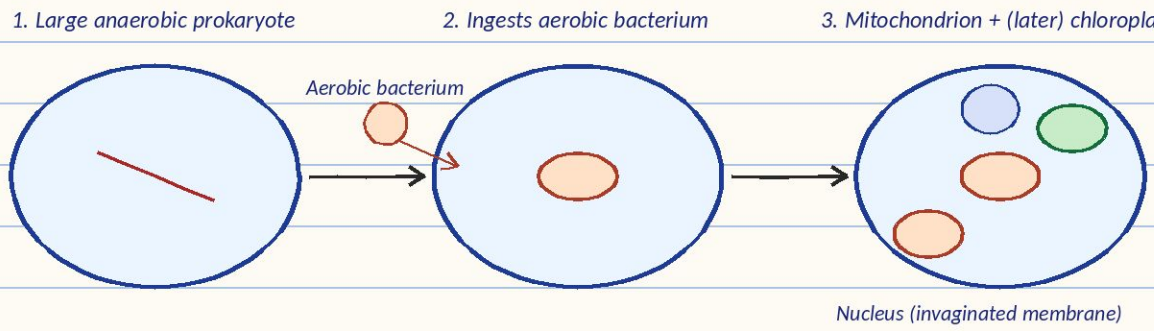
Fig: Analogous wings of bird, bat and insect

Divergent (homologous)	***Convergent (analogous)***
Common ancestor	*Different ancestors*
Different habitats	*Common habitat*
Arm of man, bat wing, whale flipper	*Wings of bird, bat, insect*

Q6. [SLO 5] Describe the endosymbiotic theory of the evolution of eukaryotes from prokaryotes.

Ans:

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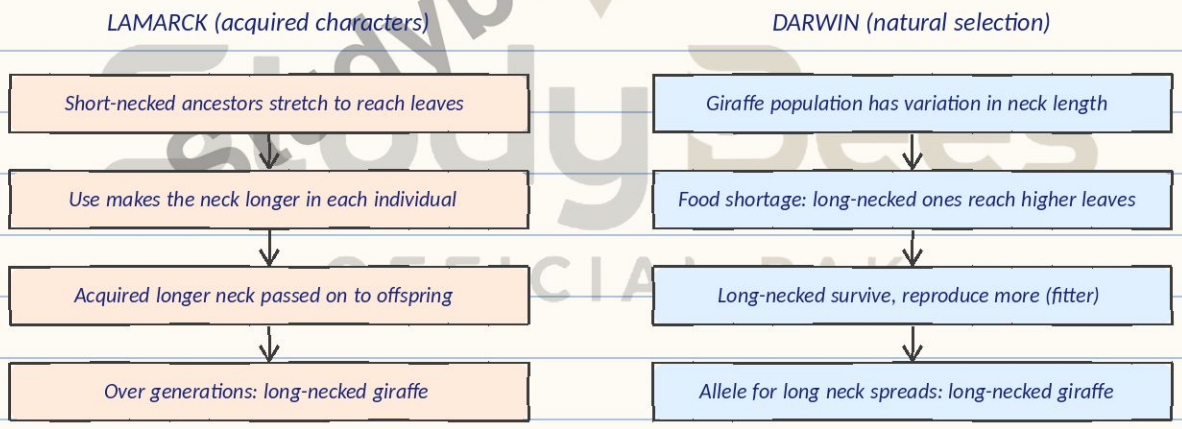
Evidence: own circular DNA, 70S ribosomes, double membrane, binary fission, size of bacteria
Spirochete ingestion may have given flagella / cilia; photosynthetic bacterium gave chloroplasts

Fig: Endosymbiotic origin of eukaryotes (Lynn Margulis)

- Lynn Margulis: large anaerobic prokaryote ingested small aerobic bacterium (became mitochondrion); a later event with a photosynthetic bacterium (chloroplast); spirochetes may have given flagella. Evidence: own circular DNA, 70S ribosomes, size and fission, double membranes.

Q7. [SLO 6, 7] Describe Lamarck's theory with the giraffe example and state its drawbacks. Compare it with Darwin's explanation.

Ans:



Lamarck: individual changes are inherited (not accepted) | Darwin: variation in population + selection

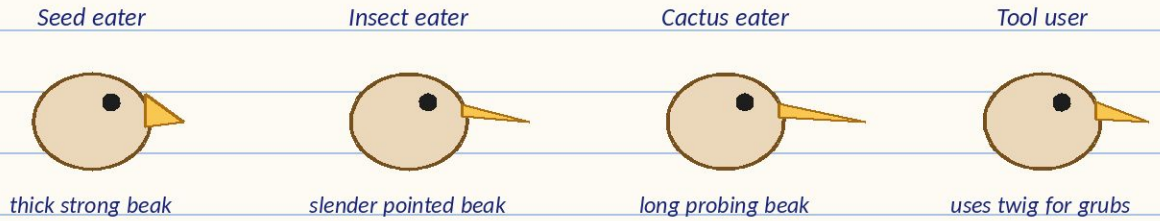
Fig: Evolution of giraffe neck - Lamarck and Darwin

- Lamarck: use and disuse; inheritance of acquired characters. Drawbacks: no proof, genotype unchanged, somatic changes not heritable, organs do not change by wish.

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Q8. [SLO 1] Describe Darwin's observations during the voyage of the Beagle and how they led to the theory of natural selection.

Ans:



Darwin's 14 finch species (Galapagos): one ancestor from the mainland, beaks adapted to different foods

Geographical barrier (sea) + different habitats: adaptive radiation and speciation

Fig: Darwin's finches (adaptive radiation)

- *South America: distinct flora and fauna; fossils resemble living species. Galapagos: 14 finches with beaks adapted to different foods. Influences: Lyell (old earth), Malthus (struggle for existence), Wallace (similar theory, 1858). Origin of Species published in 1859.*

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